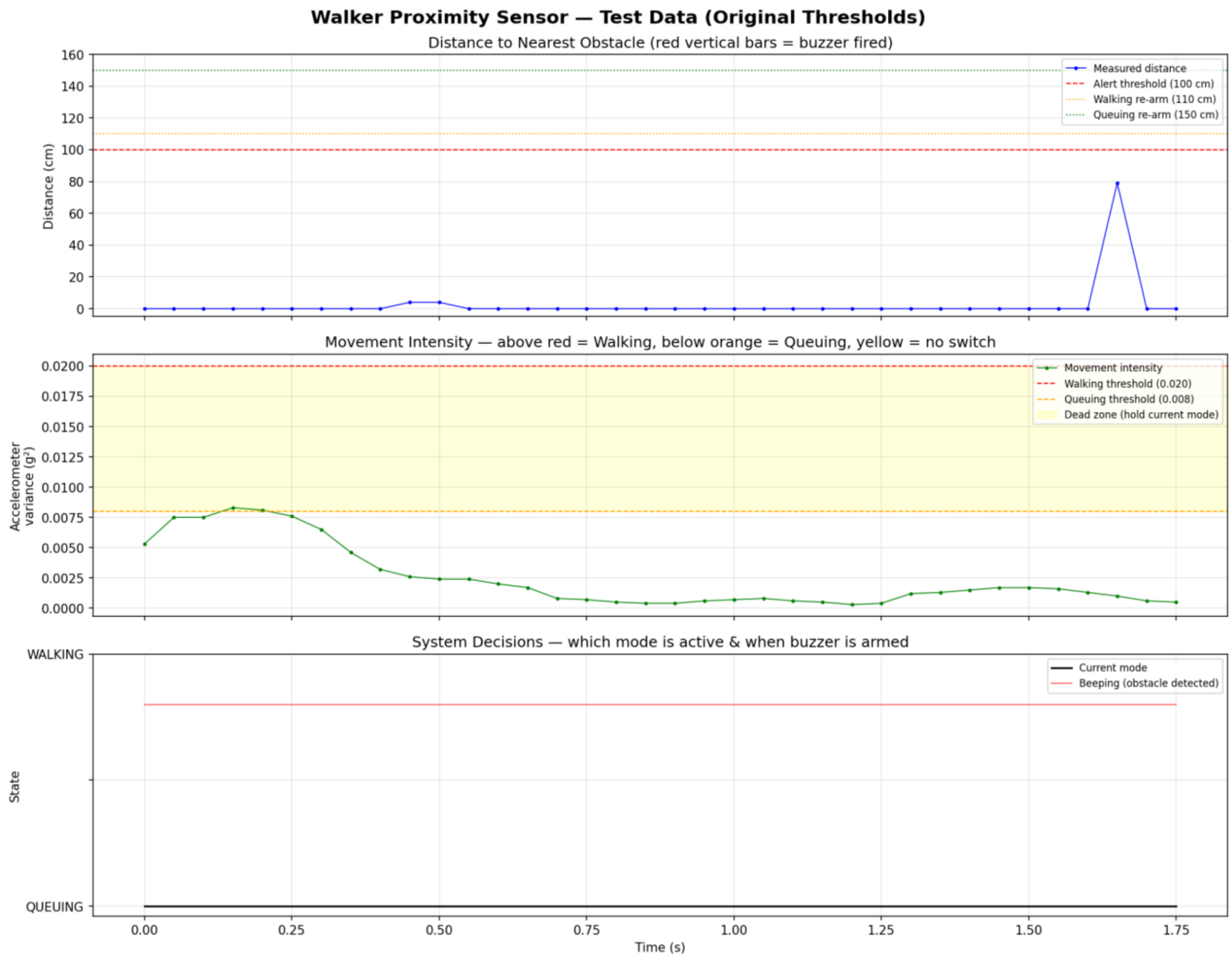
About the two plots

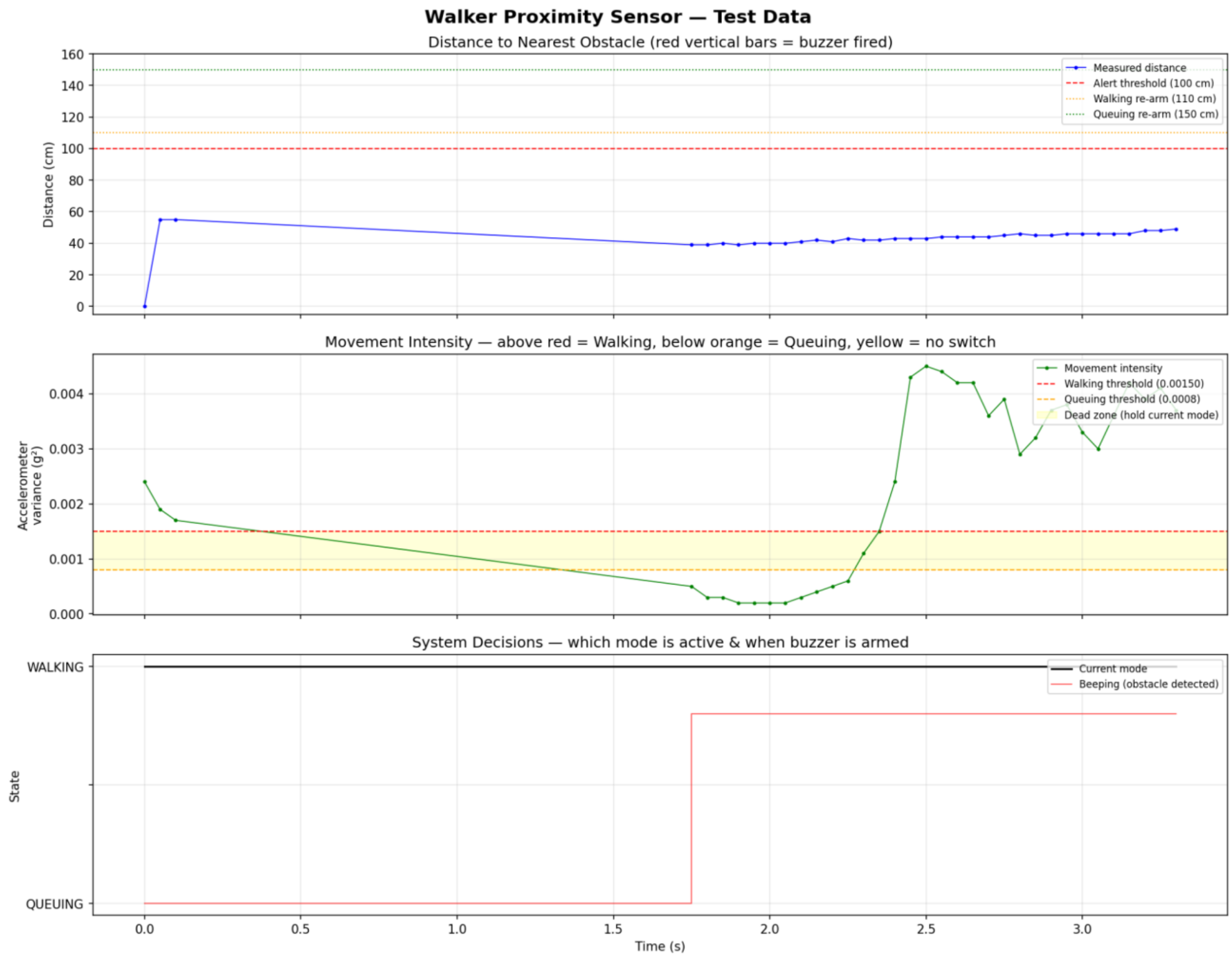
Plot A: Original thresholds. Movement intensity never reaches the Walking threshold (0.020), so the system stays stuck in Queuing mode even while the walker is moving. Threshold was too high for our hardware.

Plot B: After lowering the threshold. System correctly switches from Queuing to Walking when the walker starts moving, and Beeping triggers when an obstacle is detected. Confirms the new threshold works.

Takeaway: Default thresholds need calibration per hardware — we tune them by observing actual sensor values during testing.

Same for distance threshold testing




Walker Proximity Sensor — Test Data
Distance to Nearest Obstacle (red vertical bars = buzzer fired)
Distance (cm)
Measured distance
Alert threshold (100 cm)
Walking re-arm (110 cm)
Queuing re-arm (150 cm)
Movement Intensity — above red = Walking, below orange = Queuing, yellow = no switch
Accelerometer variance (g²)
Movement intensity
Walking threshold (0.00150)
Queuing threshold (0.0008)
Dead zone (hold current mode)
System Decisions — which mode is active & when buzzer is armed
State
WALKING
QUEUING
Current mode
Beeping (obstacle detected)
Time (s)